FrameGuard Report

October 21, 2026 at 20:10 EDT | Profile: 4K UHD @ 30fps

6 Total	1 Passed	2 Failed	3 Warnings

Summary

Filename	Codec	Resolution	Result	Summary
Interview_B-roll_proxy.mp4	h264	1920x1080	FAIL	Video resolution doesn't match the target, Frame rate doesn't match the target, Video quality (bitrate) is too low.
PROJ_CAM05_TAKE_20261018_C005.mp4	h264	3840x2160	FAIL	Video quality (bitrate) is too low.
PROJ_CAM04_TAKE_20261018_C004.mp4	h264	3840x2160	WARNING	Heads up: long stretches of silence found.
PROJ_CAM02_TAKE_20261018_C002.mov	h264	3840x2160	WARNING	Heads up: picture levels or colours are outside broadcast range.
PROJ_CAM03_TAKE_20261018_C003.mp4	h264	3840x2160	WARNING	Heads up: unexpected black frames detected.
PROJ_CAM01_TAKE_20261018_C001.mp4	h264	3840x2160	PASS	All checks passed.

Scan Specification

Resolution	3840x2160 exactly
Frame rate	30.0 ± 0.5 fps
Min video bitrate	20 Mbps
Video codecs	h264, hevc, h265, av1, prores, prores_ks, prores_aw
Containers	.mp4, .mov, .mkv, .mxf
Audio codecs	aac, pcm_s16le, pcm_s24le, pcm_s32le, pcm_s24be, pcm_s32be, pcm_f32le, opus, flac
Audio	required, min 44100 Hz
Frame timing	constant required (>1% irregular = VFR)
Scan type	progressive required
Bit depth / transfer	not enforced / not enforced
Projection / stereo	not enforced / not enforced
A/V start offset	warn beyond ±5 ms; fail beyond 40 ms early / 60 ms late
A/V end	audio ≤2 frames short, ≤2 s long
Profiles / pixel formats	any / any
Decode errors allowed	0
Black / freeze	black ≥0.3 s at ≤0.02 level on ≥98% of the picture; freeze ≥1.0 s
Picture signal	EBU R 103 levels advisory; BT.1702 flash screening on; interlace detection on
Silence	≤-50 dB for ≥2.0 s; dead at 95%
Loudness	measured and reported, not enforced

Tools: FrameGuard 0.9.1 | ffmpeg version 9.0.2-https://www.martin-riedl.de | ffprobe version 9.0.2-https://www.martin-riedl.de

Detailed Results

FAIL **Interview_B-roll_proxy.mp4**

1920x1080 | H264 | 24.00 fps | 00:00:32 | 34.1 MB

Video resolution doesn't match the target, Frame rate doesn't match the target, Video quality (bitrate) is too low.

Status	Check	Detail
PASS	Container Format	34.1 MB, container: .mp4
PASS	Video Codec	Codec: h264
FAIL	Resolution	1920x1080 is smaller than the 3840x2160 target. Conforming would upscale the image.
PASS	Aspect Ratio	1.778:1 matches the target 1.778:1
FAIL	Frame Rate	24.000 fps does not match the 30 ± 0.5 fps target.
PASS	CFR Check	Constant frame rate: 751 intervals of 41.67 ms (24.000 fps)
PASS	Scan Type	Progressive (declared progressive)
PASS	Color / HDR	8-bit, SDR (bt709), primaries bt709 (not enforced by this spec)
PASS	XR Metadata	No spherical metadata (flat), no stereo metadata (mono) (not enforced by this spec)
FAIL	Bitrate	8.7 Mbps is 56% below minimum 20 Mbps. Likely visible compression artifacts.
PASS	Audio Stream	1 audio track(s)
PASS	Audio Codec	Codec: aac
PASS	Audio Channels	2 channel(s), layout: stereo
PASS	Audio Sample Rate	48000 Hz (minimum 44100 Hz)
PASS	A/V Duration	Video: 32.00s, Audio: 32.00s (diff: 0.000s)
PASS	Frame Count	768 frames (expected ~768, diff: 0)
PASS	Codec Profile	H.264 Constrained Baseline @ Level 4.1, yuv420p (Level 4 required)
PASS	Container Structure	ISO-BMFF, 2 audio/video track(s), every box and sample inside the file; index (moov) after the media data, not fast-start (fine for file delivery); brand 'isom'.
PASS	Edit List	Video: one edit starting at media tick 0; audio: one edit starting at media tick 1024 (encoder priming). Normal.
PASS	A/V Sync Offset	Audio and video start together (+0.0 ms; container timing only)
PASS	Track Layout	1 video, 1 audio
PASS	File Integrity	Software decode of 768/768 video frames, 32.00/32.00 s audio; 0 decoder error(s).
PASS	Black Frames	No black segments detected.
PASS	Freeze Frames	No freezes ≥1s detected.
PASS	Signal Levels	Largest share outside EBU R 103 (limit 1% of the picture): luma max 0.00%, RGB gamut max 0.07%.
PASS	Flash Risk	No flashing above BT.1702-3 limits detected (ITU-R BT.1702-3 screening only; PSE certification requires a Harding FPA class analyser).
PASS	Interlace Detection	No interlace combing found (0% of frames look interlaced).

Interview_B-roll_proxy.mp4 (continued)

Status	Check	Detail
PASS	Audio Silence	No extended silence detected.
PASS	Dead Channel	All 2 channels carry signal.
PASS	Audio Loudness	-18.7 LUFS integrated, LRA 0.0 LU; not enforced (EBU R 128 target -23 ± 0.5 LU)
PASS	Audio True Peak	true peak -16.3 dBTP, sample peak -16.3 dBFS; not enforced (EBU R 128 max -1 dBTP)
PASS	Audio Clipping	No clipped runs (≥3 samples at full scale or at the channel peak).
PASS	Audio Dropouts	No dropouts to ≤ -90 dBFS between 5 ms and 2 s.
PASS	Audio Phase	L/R correlation +0.00, never below -0.2 for 1 s or more.
PASS	Dual Mono	L/R distinct (correlation 0.000).
PASS	DC Offset	Largest DC offset none (limit -50 dBFS).
PASS	Audio Hum	No steady 50/60 Hz tone (50 Hz -114.1 dBFS, 60 Hz -113.5 dBFS; floor -65 dBFS).

FAIL PROJ_CAM05_TAKE_20261018_C005.mp4

3840x2160 | H264 | 30.00 fps | 00:00:32 | 36.9 MB

Video quality (bitrate) is too low.

Status	Check	Detail
PASS	Container Format	36.9 MB, container: .mp4
PASS	Video Codec	Codec: h264
PASS	Resolution	3840x2160 (matches 3840x2160 target)
PASS	Aspect Ratio	1.778:1 matches the target 1.778:1
PASS	Frame Rate	30.000 fps (matches the 30 fps target)
PASS	CFR Check	Constant frame rate: 943 intervals of 33.33 ms (30.000 fps)
PASS	Scan Type	Progressive (declared progressive)
PASS	Color / HDR	8-bit, SDR (bt709), primaries bt709 (not enforced by this spec)
PASS	XR Metadata	No spherical metadata (flat), no stereo metadata (mono) (not enforced by this spec)
FAIL	Bitrate	9.5 Mbps is 53% below minimum 20 Mbps. Likely visible compression artifacts.
PASS	Audio Stream	1 audio track(s)
PASS	Audio Codec	Codec: aac
PASS	Audio Channels	2 channel(s), layout: stereo
PASS	Audio Sample Rate	48000 Hz (minimum 44100 Hz)
PASS	A/V Duration	Video: 32.00s, Audio: 32.00s (diff: 0.000s)
PASS	Frame Count	960 frames (expected ~960, diff: 0)
PASS	Codec Profile	H.264 Constrained Baseline @ Level 5.1, yuv420p (Level 5.1 required)
PASS	Container Structure	ISO-BMFF, 2 audio/video track(s), every box and sample inside the file; index (moov) after the media data, not fast-start (fine for file delivery); brand 'isom'.

PROJ_CAM05_TAKE_20261018_C005.mp4 (continued)

Status	Check	Detail
PASS	Edit List	Video: one edit starting at media tick 0; audio: one edit starting at media tick 1024 (encoder priming). Normal.
PASS	A/V Sync Offset	Audio and video start together (+0.0 ms; container timing only)
PASS	Track Layout	1 video, 1 audio
PASS	File Integrity	Software decode of 960/960 video frames, 32.00/32.00 s audio; 0 decoder error(s).
PASS	Black Frames	No black segments detected.
PASS	Freeze Frames	No freezes ≥1s detected.
PASS	Signal Levels	Largest share outside EBU R 103 (limit 1% of the picture): luma max 0.00%, RGB gamut max 0.10%.
PASS	Flash Risk	No flashing above BT.1702-3 limits detected (ITU-R BT.1702-3 screening only; PSE certification requires a Harding FPA class analyser).
PASS	Interlace Detection	No interlace combing found (0% of frames look interlaced).
PASS	Audio Silence	No extended silence detected.
PASS	Dead Channel	All 2 channels carry signal.
PASS	Audio Loudness	-18.7 LUFS integrated, LRA 0.0 LU; not enforced (EBU R 128 target -23 ± 0.5 LU)
PASS	Audio True Peak	true peak -16.3 dBTP, sample peak -16.3 dBFS; not enforced (EBU R 128 max -1 dBTP)
PASS	Audio Clipping	No clipped runs (≥3 samples at full scale or at the channel peak).
PASS	Audio Dropouts	No dropouts to ≤ -90 dBFS between 5 ms and 2 s.
PASS	Audio Phase	L/R correlation +0.00, never below -0.2 for 1 s or more.
PASS	Dual Mono	L/R distinct (correlation 0.000).
PASS	DC Offset	Largest DC offset none (limit -50 dBFS).
PASS	Audio Hum	No steady 50/60 Hz tone (50 Hz -114.1 dBFS, 60 Hz -113.5 dBFS; floor -65 dBFS).

WARNING PROJ_CAM04_TAKE_20261018_C004.mp4

3840x2160 | H264 | 30.00 fps | 00:00:32 | 109.5 MB

Heads up: long stretches of silence found.

Status	Check	Detail
PASS	Container Format	109.5 MB, container: .mp4
PASS	Video Codec	Codec: h264
PASS	Resolution	3840x2160 (matches 3840x2160 target)
PASS	Aspect Ratio	1.778:1 matches the target 1.778:1
PASS	Frame Rate	30.000 fps (matches the 30 fps target)
PASS	CFR Check	Constant frame rate: 943 intervals of 33.33 ms (30.000 fps)
PASS	Scan Type	Progressive (declared progressive)

PROJ_CAM04_TAKE_20261018_C004.mp4 (continued)

Status	Check	Detail
PASS	Color / HDR	8-bit, SDR (bt709), primaries bt709 (not enforced by this spec)
PASS	XR Metadata	No spherical metadata (flat), no stereo metadata (mono) (not enforced by this spec)
PASS	Bitrate	28.5 Mbps (min: 20 Mbps)
PASS	Audio Stream	1 audio track(s)
PASS	Audio Codec	Codec: aac
PASS	Audio Channels	2 channel(s), layout: stereo
PASS	Audio Sample Rate	48000 Hz (minimum 44100 Hz)
PASS	A/V Duration	Video: 32.00s, Audio: 32.00s (diff: 0.000s)
PASS	Frame Count	960 frames (expected ~960, diff: 0)
PASS	Codec Profile	H.264 Constrained Baseline @ Level 5.1, yuv420p (Level 5.1 required)
PASS	Container Structure	ISO-BMFF, 2 audio/video track(s), every box and sample inside the file; index (moov) after the media data, not fast-start (fine for file delivery); brand 'isom'.
PASS	Edit List	Video: one edit starting at media tick 0; audio: one edit starting at media tick 1024 (encoder priming). Normal.
PASS	A/V Sync Offset	Audio and video start together (+0.0 ms; container timing only)
PASS	Track Layout	1 video, 1 audio
PASS	File Integrity	Software decode of 960/960 video frames, 32.00/32.00 s audio; 0 decoder error(s).
PASS	Black Frames	No black segments detected.
PASS	Freeze Frames	No freezes ≥1s detected.
PASS	Signal Levels	Largest share outside EBU R 103 (limit 1% of the picture): luma max 0.00%, RGB gamut max 0.01%.
PASS	Flash Risk	No flashing above BT.1702-3 limits detected (ITU-R BT.1702-3 screening only; PSE certification requires a Harding FPA class analyser).
PASS	Interlace Detection	No interlace combing found (0% of frames look interlaced).
WARN	Audio Silence	1 silence region(s), total 3.5s. At: 00:00:10.008 (3.5s)
PASS	Dead Channel	All 2 channels carry signal.
PASS	Audio Loudness	-18.7 LUFS integrated, LRA 3.0 LU; not enforced (EBU R 128 target -23 ± 0.5 LU)
PASS	Audio True Peak	true peak -16.3 dBTP, sample peak -16.3 dBFS; not enforced (EBU R 128 max -1 dBTP)
PASS	Audio Clipping	No clipped runs (≥3 samples at full scale or at the channel peak).
PASS	Audio Dropouts	No dropouts to ≤ -90 dBFS between 5 ms and 2 s.
PASS	Audio Phase	L/R correlation +0.00, never below -0.2 for 1 s or more.
PASS	Dual Mono	L/R distinct (correlation 0.000).
PASS	DC Offset	Largest DC offset -120.0 dBFS (limit -50 dBFS).
PASS	Audio Hum	No steady 50/60 Hz tone (50 Hz -101.1 dBFS, 60 Hz -100.9 dBFS; floor -65 dBFS).

WARNING

PROJ_CAM02_TAKE_20261018_C002.mov

3840x2160 | H264 | 30.00 fps | 00:00:32 | 109.9 MB

Heads up: picture levels or colours are outside broadcast range.

Status	Check	Detail
PASS	Container Format	109.9 MB, container: .mov
PASS	Video Codec	Codec: h264
PASS	Resolution	3840x2160 (matches 3840x2160 target)
PASS	Aspect Ratio	1.778:1 matches the target 1.778:1
PASS	Frame Rate	30.000 fps (matches the 30 fps target)
PASS	CFR Check	Constant frame rate: 943 intervals of 33.33 ms (30.000 fps)
PASS	Scan Type	Progressive (declared progressive)
PASS	Color / HDR	8-bit, SDR (bt709), primaries bt709 (not enforced by this spec)
PASS	XR Metadata	No spherical metadata (flat), no stereo metadata (mono) (not enforced by this spec)
PASS	Bitrate	28.6 Mbps (min: 20 Mbps)
PASS	Audio Stream	1 audio track(s)
PASS	Audio Codec	Codec: aac
PASS	Audio Channels	2 channel(s), layout: stereo
PASS	Audio Sample Rate	48000 Hz (minimum 44100 Hz)
PASS	A/V Duration	Video: 32.00s, Audio: 32.00s (diff: 0.000s)
PASS	Frame Count	960 frames (expected ~960, diff: 0)
PASS	Codec Profile	H.264 Constrained Baseline @ Level 5.1, yuv420p (Level 5.1 required)
PASS	Container Structure	ISO-BMFF, 2 audio/video track(s), every box and sample inside the file; index (moov) after the media data, not fast-start (fine for file delivery); brand 'qt'.
PASS	Edit List	Video: one edit starting at media tick 0; audio: one edit starting at media tick 1024 (encoder priming). Normal.
PASS	A/V Sync Offset	Audio and video start together (+0.0 ms; container timing only)
PASS	Track Layout	1 video, 1 audio
PASS	File Integrity	Software decode of 960/960 video frames, 32.00/32.00 s audio; 0 decoder error(s).
PASS	Black Frames	No black segments detected.
PASS	Freeze Frames	No freezes ≥1s detected.
WARN	Signal Levels	RGB gamut outside the R 103 range on >1% of the picture: 00:00:00.000 (32.0s) up to 97.2%. (Largest share outside EBU R 103 (limit 1% of the picture): luma max 0.00%, RGB gamut max 97.23%.)
PASS	Flash Risk	No flashing above BT.1702-3 limits detected (ITU-R BT.1702-3 screening only; PSE certification requires a Harding FPA class analyser).
PASS	Interlace Detection	No interlace combing found (0% of frames look interlaced).
PASS	Audio Silence	No extended silence detected.
PASS	Dead Channel	All 2 channels carry signal.

PROJ_CAM02_TAKE_20261018_C002.mov (continued)

Status	Check	Detail
PASS	Audio Loudness	-18.7 LUFS integrated, LRA 0.0 LU; not enforced (EBU R 128 target -23 ± 0.5 LU)
PASS	Audio True Peak	true peak -16.3 dBTP, sample peak -16.3 dBFS; not enforced (EBU R 128 max -1 dBTP)
PASS	Audio Clipping	No clipped runs (≥3 samples at full scale or at the channel peak).
PASS	Audio Dropouts	No dropouts to ≤ -90 dBFS between 5 ms and 2 s.
PASS	Audio Phase	L/R correlation +0.00, never below -0.2 for 1 s or more.
PASS	Dual Mono	L/R distinct (correlation 0.000).
PASS	DC Offset	Largest DC offset none (limit -50 dBFS).
PASS	Audio Hum	No steady 50/60 Hz tone (50 Hz -114.1 dBFS, 60 Hz -113.5 dBFS; floor -65 dBFS).

WARNING PROJ_CAM03_TAKE_20261018_C003.mp4

3840x2160 | H264 | 30.00 fps | 00:00:32 | 104.7 MB

Heads up: unexpected black frames detected.

Status	Check	Detail
PASS	Container Format	104.7 MB, container: .mp4
PASS	Video Codec	Codec: h264
PASS	Resolution	3840x2160 (matches 3840x2160 target)
PASS	Aspect Ratio	1.778:1 matches the target 1.778:1
PASS	Frame Rate	30.000 fps (matches the 30 fps target)
PASS	CFR Check	Constant frame rate: 943 intervals of 33.33 ms (30.000 fps)
PASS	Scan Type	Progressive (declared progressive)
PASS	Color / HDR	8-bit, SDR (bt709), primaries bt709 (not enforced by this spec)
PASS	XR Metadata	No spherical metadata (flat), no stereo metadata (mono) (not enforced by this spec)
PASS	Bitrate	27.2 Mbps (min: 20 Mbps)
PASS	Audio Stream	1 audio track(s)
PASS	Audio Codec	Codec: aac
PASS	Audio Channels	2 channel(s), layout: stereo
PASS	Audio Sample Rate	48000 Hz (minimum 44100 Hz)
PASS	A/V Duration	Video: 32.00s, Audio: 32.00s (diff: 0.000s)
PASS	Frame Count	960 frames (expected ~960, diff: 0)
PASS	Codec Profile	H.264 Constrained Baseline @ Level 5.1, yuv420p (Level 5.1 required)
PASS	Container Structure	ISO-BMFF, 2 audio/video track(s), every box and sample inside the file; index (moov) after the media data, not fast-start (fine for file delivery); brand 'isom'.
PASS	Edit List	Video: one edit starting at media tick 0; audio: one edit starting at media tick 1024 (encoder priming). Normal.

PROJ_CAM03_TAKE_20261018_C003.mp4 (continued)

Status	Check	Detail
PASS	A/V Sync Offset	Audio and video start together (+0.0 ms; container timing only)
PASS	Track Layout	1 video, 1 audio
PASS	File Integrity	Software decode of 960/960 video frames, 32.00/32.00 s audio; 0 decoder error(s).
WARN	Black Frames	1 black segment(s) ≥0.3s, total 2.5s (≥98% of the picture at black level). At: 00:00:14.000 (2.5s).
PASS	Freeze Frames	No freezes ≥1s detected.
PASS	Signal Levels	Largest share outside EBU R 103 (limit 1% of the picture): luma max 0.00%, RGB gamut max 0.01%.
PASS	Flash Risk	No flashing above BT.1702-3 limits detected (ITU-R BT.1702-3 screening only; PSE certification requires a Harding FPA class analyser).
PASS	Interlace Detection	No interlace combing found (0% of frames look interlaced).
PASS	Audio Silence	No extended silence detected.
PASS	Dead Channel	All 2 channels carry signal.
PASS	Audio Loudness	-18.7 LUFS integrated, LRA 0.0 LU; not enforced (EBU R 128 target -23 ± 0.5 LU)
PASS	Audio True Peak	true peak -16.3 dBTP, sample peak -16.3 dBFS; not enforced (EBU R 128 max -1 dBTP)
PASS	Audio Clipping	No clipped runs (≥3 samples at full scale or at the channel peak).
PASS	Audio Dropouts	No dropouts to ≤ -90 dBFS between 5 ms and 2 s.
PASS	Audio Phase	L/R correlation +0.00, never below -0.2 for 1 s or more.
PASS	Dual Mono	L/R distinct (correlation 0.000).
PASS	DC Offset	Largest DC offset none (limit -50 dBFS).
PASS	Audio Hum	No steady 50/60 Hz tone (50 Hz -114.1 dBFS, 60 Hz -113.5 dBFS; floor -65 dBFS).

PASS PROJ_CAM01_TAKE_20261018_C001.mp4

3840x2160 | H264 | 30.00 fps | 00:00:32 | 109.7 MB

Status	Check	Detail
PASS	Container Format	109.7 MB, container: .mp4
PASS	Video Codec	Codec: h264
PASS	Resolution	3840x2160 (matches 3840x2160 target)
PASS	Aspect Ratio	1.778:1 matches the target 1.778:1
PASS	Frame Rate	30.000 fps (matches the 30 fps target)
PASS	CFR Check	Constant frame rate: 943 intervals of 33.33 ms (30.000 fps)
PASS	Scan Type	Progressive (declared progressive)
PASS	Color / HDR	8-bit, SDR (bt709), primaries bt709 (not enforced by this spec)
PASS	XR Metadata	No spherical metadata (flat), no stereo metadata (mono) (not enforced by this spec)

PROJ_CAM01_TAKE_20261018_C001.mp4 (continued)

Status	Check	Detail
PASS	**Bitrate**	28.6 Mbps (min: 20 Mbps)
PASS	**Audio Stream**	1 audio track(s)
PASS	**Audio Codec**	Codec: aac
PASS	**Audio Channels**	2 channel(s), layout: stereo
PASS	**Audio Sample Rate**	48000 Hz (minimum 44100 Hz)
PASS	**A/V Duration**	Video: 32.00s, Audio: 32.00s (diff: 0.000s)
PASS	**Frame Count**	960 frames (expected ~960, diff: 0)
PASS	**Codec Profile**	H.264 Constrained Baseline @ Level 5.1, yuv420p (Level 5.1 required)
PASS	**Container Structure**	ISO-BMFF, 2 audio/video track(s), every box and sample inside the file; index (moov) after the media data, not fast-start (fine for file delivery); brand 'isom'.
PASS	**Edit List**	Video: one edit starting at media tick 0; audio: one edit starting at media tick 1024 (encoder priming). Normal.
PASS	**A/V Sync Offset**	Audio and video start together (+0.0 ms; container timing only)
PASS	**Track Layout**	1 video, 1 audio
PASS	**File Integrity**	Software decode of 960/960 video frames, 32.00/32.00 s audio; 0 decoder error(s).
PASS	**Black Frames**	No black segments detected.
PASS	**Freeze Frames**	No freezes ≥1s detected.
PASS	**Signal Levels**	Largest share outside EBU R 103 (limit 1% of the picture): luma max 0.00%, RGB gamut max 0.01%.
PASS	**Flash Risk**	No flashing above BT.1702-3 limits detected (ITU-R BT.1702-3 screening only; PSE certification requires a Harding FPA class analyser).
PASS	**Interlace Detection**	No interlace combing found (0% of frames look interlaced).
PASS	**Audio Silence**	No extended silence detected.
PASS	**Dead Channel**	All 2 channels carry signal.
PASS	**Audio Loudness**	-18.7 LUFS integrated, LRA 0.0 LU; not enforced (EBU R 128 target -23 ± 0.5 LU)
PASS	**Audio True Peak**	true peak -16.3 dBTP, sample peak -16.3 dBFS; not enforced (EBU R 128 max -1 dBTP)
PASS	**Audio Clipping**	No clipped runs (≥3 samples at full scale or at the channel peak).
PASS	**Audio Dropouts**	No dropouts to ≤ -90 dBFS between 5 ms and 2 s.
PASS	**Audio Phase**	L/R correlation +0.00, never below -0.2 for 1 s or more.
PASS	**Dual Mono**	L/R distinct (correlation 0.000).
PASS	**DC Offset**	Largest DC offset none (limit -50 dBFS).
PASS	**Audio Hum**	No steady 50/60 Hz tone (50 Hz -114.1 dBFS, 60 Hz -113.5 dBFS; floor -65 dBFS).